Linear System — Homework 4

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Exercise 1

Determine whether or not the following matrix is nondefective.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

Solution

$$\begin{aligned} \det(A - sI) &= \begin{bmatrix} -s & 1 & 0 \\ 0 & -s & 1 \\ -1 & 1 & 1-s \end{bmatrix} \\ &= -s\Big(s(s-1) - 1\Big) + (-1) \cdot 1 \\ &= -s(s^2 - s - 1) - 1 \\ &= -s^3 + s^2 + s - 1 \\ &= -(s-1)^2(s+1) \end{aligned}$$

for $s = 1$

$$A - sI = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \Rightarrow \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Hence, for eigenvalue $s = 1$, AM $= 2$, and GM $= 1$

for $s = -1$

$$A - sI = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \Rightarrow \mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Hence, for eigenvalue $s = -1$, AM $= 1$, and GM $= 1$

Therefore, matrix A is defective.

Exercise 2

What is the companion matrix whose eigenvalues are -1, -2, and -3?

Solution

$$\begin{aligned} p(s) &= (s+1)(s+2)(s+3) \\ &= a_3 s^3 + a_2 s^2 + a_1 s + a_0 \\ &= s^3 + 6s^2 + 11s + 6 \end{aligned}$$

Hence, companion matrix is

$$\begin{aligned} A &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \end{aligned}$$

Exercise 3

What is the real 2×2 companion matrix with eigenvalues $1 + j$, $1 - j$?

Solution

$$\begin{aligned} p(s) &= (s - (1+j))(s - (1-j)) \\ &= a_2 s^2 + a_1 s + a_0 \\ &= s^2 - 2s + 2 \end{aligned}$$

Hence, companion matrix is

$$\begin{aligned} A &= \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ -2 & 2 \end{bmatrix} \end{aligned}$$

Exercise 4

Suppose A is a real square matrix and the vectors

$$\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \qquad \begin{pmatrix} 1 \\ 2 \\ 4 \\ 8 \end{pmatrix}, \qquad \begin{pmatrix} 1 \\ j \\ -1 \\ -j \end{pmatrix}$$

are eigenvectors of A corresponding to eigenvalues -1, 2, and j, respectively. What is the response $x(t)$ of the system

$$\dot{x} = Ax$$

to the following initial conditions?

(a) $x(0) = \begin{pmatrix} 2 \\ -2 \\ 2 \\ -2 \end{pmatrix}$

Solution

Since $\mathbf{v}_1$ is an eigenvector corresponding to eigenvalue $\lambda_1 = -1$,

$$x(0) = \begin{bmatrix} 2 \\ -2 \\ 2 \\ -2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = 2\mathbf{v}_1 \triangleq \mathbf{v}$$

$$A\mathbf{v} = \lambda\mathbf{v} \Leftrightarrow x(t) = e^{\lambda t}\mathbf{v} = e^{\lambda t}x(0)$$

$$\begin{aligned} x(t) &= e^{\lambda_1 t}x(0) \\ &= e^{\lambda_1 t}\mathbf{v} \\ &= e^{-t} \cdot 2\mathbf{v}_1 \\ &= \begin{bmatrix} 2e^{-t} \\ -2e^{-t} \\ 2e^{-t} \\ -2e^{-t} \end{bmatrix}. \end{aligned}$$

(b) $x(0) = \begin{pmatrix} -1 \\ -2 \\ -4 \\ -8 \end{pmatrix}$

Solution

$$x(0) = \begin{bmatrix} -1 \\ -2 \\ -4 \\ -8 \end{bmatrix} = - \begin{bmatrix} 1 \\ 2 \\ 4 \\ 8 \end{bmatrix} = -\mathbf{v}_2 \text{ which corresponding to eigenvalue is } 2$$

$$A\mathbf{v} = \lambda\mathbf{v} \Leftrightarrow x(t) = e^{\lambda t}x(0) = e^{\lambda t}\mathbf{v}$$

$$\begin{aligned} x(t) &= e^{\lambda t}x(0) \\ &= e^{\lambda t}\mathbf{v} \\ &= e^{\lambda t} \cdot -\mathbf{v}_2 \\ &= e^{2t}\begin{bmatrix} -1 \\ -2 \\ -4 \\ -8 \end{bmatrix} = \begin{bmatrix} -e^{2t} \\ -2e^{2t} \\ -4e^{2t} \\ -8e^{2t} \end{bmatrix} \end{aligned}$$

(c) $x(0) = \begin{pmatrix} 0 \\ 3 \\ 3 \\ 9 \end{pmatrix}$

Solution

$$x(0) = \mathbf{v} = \begin{bmatrix} 0 \\ 3 \\ 3 \\ 9 \end{bmatrix} = -\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 4 \\ 8 \end{bmatrix} = -\mathbf{v}_1 + \mathbf{v}_2$$

with linear combination

$$\begin{aligned} x(t) &= -e^{\lambda_1 t}\mathbf{v}_1 + e^{\lambda_2 t}\mathbf{v}_2 \\ &= \begin{bmatrix} -e^{-t} + e^{2t} \\ e^{-t} + 2e^{2t} \\ -e^{-t} + 4e^{2t} \\ e^{-t} + 8e^{2t} \end{bmatrix} \end{aligned}$$

(d) $x(0) = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$

Solution

$$\mathbf{v}_3 = \begin{bmatrix} 1 \\ j \\ -1 \\ j \end{bmatrix} = \mathbf{u}_3 + j\mathbf{p}_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} + j\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \mathbf{u}_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{p}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda_3 = \alpha + j\omega = j \Rightarrow \alpha = 0, \quad \omega = 1$$

$$x(t) =$$

Exercise 5

Consider a discrete-time LTI system described by

$$x(k+1) = Ax(k).$$

(a) Suppose that the vectors

$$\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

are eigenvectors of A corresponding to eigenvalues -2 and 3, respectively. What is the response $x(k)$ of the system to the initial condition

$$x(0) = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}?$$

Solution

(b) Suppose A is a real matrix and the vector

$$\begin{pmatrix} 1 \\ j \\ -1 \\ -j \end{pmatrix}$$

is an eigenvector of A corresponding to the eigenvalue $2 + 3j$. What is the response $x(k)$ of the system to the initial condition

$$x(0) = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}?$$

Solution

Exercise 6

Consider a 5×5 matrix

$$A = vw^T$$

where v and w are 5×1 matrices.

(a) What is the rank of A?

Solution

(b) What is the nullity of A?

Solution

(c) What are the eigenvalues of A if $w^T v \neq 0$?

Solution

For Exercise 7 and 8, please use Matlab/Python to illustrate the time history plot. Please don't plot y, θ_1, θ_2 in the same plot, separate them!

Exercise 7

Recall the 2 pendulum cart system.

$$\begin{aligned}(m_0+m_1+m_2)\ddot{y} - m_1 l\cos\theta_1\,\ddot{\theta}_1 - m_2 l\cos\theta_2\,\ddot{\theta}_2 \\ + m_1 l\sin\theta_1\,\dot{\theta}_1^2 + m_2 l\sin\theta_2\,\dot{\theta}_2^2 = u \\ - m_1 l\cos\theta_1\,\ddot{y} + m_1 l^2\ddot{\theta}_1 + m_1 lg\sin\theta_1 = 0 \\ - m_2 l\cos\theta_2\,\ddot{y} + m_2 l^2\ddot{\theta}_2 + m_2 lg\sin\theta_2 = 0\end{aligned}$$

Consider the equilibrium configurations defined by $u^e = 0$ and

$$E1: \qquad (y^e, \theta_1^e, \theta_2^e) = (0,0,0)$$

and

$$E2: \qquad (y^e, \theta_1^e, \theta_2^e) = (0,\pi,\pi).$$

Consider state space representations of the linearizations corresponding to the following combinations of parameters and equilibrium conditions:

	Parameter	Equilibrium
$L1$	$P1$	$E1$
$L2$	$P1$	$E2$
$L3$	$P2$	$E1$
$L4$	$P2$	$E2$
$L5$	$P3$	$E1$
$L6$	$P3$	$E2$
$L7$	$P4$	$E1$
$L8$	$P4$	$E2$

(a) Determine the eigenvalues of all the ***linearized systems*** L1–L8.

Solution

(b) Compare the behavior of the nonlinear system with that of the linearized system for cases L7 and L8. Illustrate your results using different initial condition with time histories of y, θ_1, and θ_2. Plesae write paragraph to analysis the comparison between nonlinear and linearized system.

Initial conditions for case L7: ± 0 deg, ± 1 deg, ± 10 deg, ± 90 deg, ± 180 deg.

Initial conditions for case L8: ± 1 deg.

Solution

Exercise 8

This exercise refers to linearizations L7 and L8 of the two pendulum cart system.

(a) For L7 choose an initial state for the linearized system which results in a periodic solution for the linearized system.

Solution

(b) For L8 choose an initial state for the linearized system which results in a solution which asymptotically decays to zero for the linearized system.

Solution

(c) For L8 choose an initial state for the linearized system which results in a solution whose magnitude grows exponentially for the linearized system.

Solution

In each case, simulate both the linearized system and the nonlinear system with initial conditions corresponding to your chosen initial state for the linearized system.